Concepts of Programming Languages

Stateful Languages

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Outline

- State - in Scala and in our languages
- An interpretation style for modeling the effect of state
- Variables
- Mutable data structures (Boxes)
- Summary

Variables and Mutable Data Structures

Many languages, e.g., C/C++, Java, Scala, etc., have mutable data structures

Two kinds of mutations:

- mutating the value bound to an identifier
- mutating a value in a container

Variables and Mutable Data Structures

Mutating an identifier

```
int i;
i = nextPrime(12)
i = nextPrime(18)
```

- `i` must literally be an identifier.
- Identifiers with changeable values are called **variables**.

Mutating a value in a container

`o.f = e`	(object in Java)
`list.add(e)`	(collection in Java)

Containers change their internal state but do not immediately reflect this change to the outside.

State in Scala

- Scala also has stateful constructs: variables and boxes

```
var v = 1
while (v < 100) { v = v * 3 }

object o {
  var f: Int
}
o.f = 17
```

- a variable can be ***assigned*** different values over time
- a box can ***contain*** different values over time
- a box can be passed around to allow mutation by others

```
def foo(o: o.type) {
  o.f = 13;
}
```

This Lecture

We will implement both variables and boxes with their operations to understand (a) their differences and (b) interactions with other language features.

We will also implement sequencing.

Is Mutable State Essential?

Use of State is not Essential

```
C:

int i;
sum = 0;
for (i=0; i < 100; i++) {
  sum += f(i);
}
```

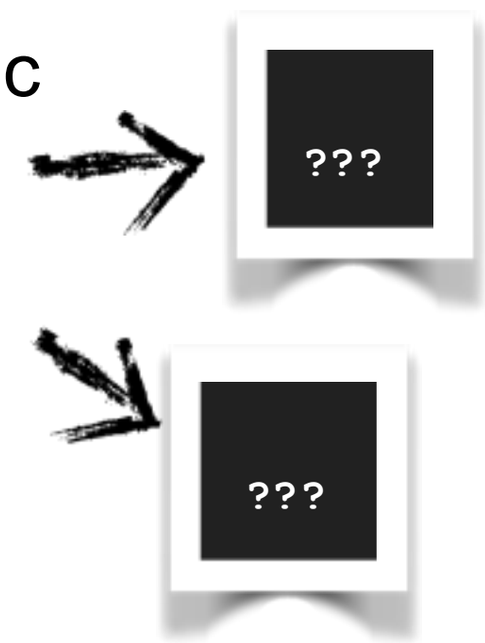
```
Haskell:

foldr (+) 0 (map f [0..99])
```

these two code snippets do the same thing

State Complicates Reasoning About Programs

Requires to reason differently about program before and after every mutation.

```
b = 2
c = 3
a = b + c
print(a)
c = 5
print(a)
```



State Complicates Reasoning About Programs

Requires to reason differently about program before and after every mutation.

```
b = 2
c = 3
a = b + c
print(a)  // 5
c = 5
print(a)  // 7
```

The same instruction produces different results at diffrent points in execution time

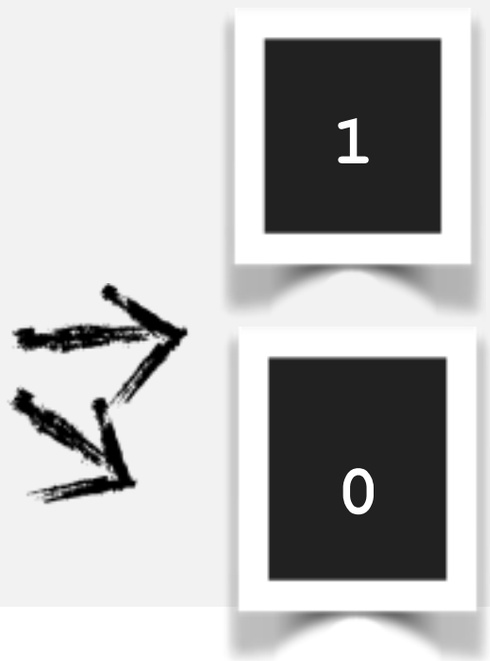
State Complicates Reasoning About Programs

```
Let("switch", 0,
  Let("toggle",
    Fun("_",
      If0(
        "switch",
        Seqn(
          SetId("switch", 1),
          1),
        Seqn(
          SetId("switch", 0),
          0))),

    Add(
      App("toggle", 42),
      App("toggle", 42))
  )
)
```

What is the result of evaluating the two operands of +

State Complicates Reasoning About Programs

```
Let("switch", 0,
  Let("toggle",
    Fun("_",
      If0(
        "switch",
        Seqn(
          SetId("switch", 1),
          1),
        Seqn(
          SetId("switch", 0),
          0))),

    Add(
      App("toggle", 42),
      App("toggle", 42))
  )
)
```



Same expression reduced to different values at two different places in the program (at two different times during the execution. Why?)

State May Cause Problems

Easily to cause bugs in multi-threaded environments.

```
# Thread 1
x = x + 1

# Thread 2
x = x + 1
```

What should x be after both threads run if it started at 0?

Multiple threads can access and modify the same variable (shared state) at the same time.

If operations are not coordinated, the outcome depends on timing, leading to unpredictable bugs known as race conditions

However ... Legitimate Uses of State

- In the real world, there are events that truly alter the state.
- Cars consume fuel, money is deposited or withdrawn from bank-accounts, ...

- Copying data structures is not a good fit in these cases.
- A mutable data structure might be a better solution.

Our starting point: CFLAE

```
<CFLAE> ::= <num>
          | {+ <CFLAE> <CFLAE>}
          | {- <CFLAE> <CFLAE>}
          | {* <CFLAE> <CFLAE>}
          | {let {<id> <CFLAE>} <CFLAE>}
          | <id>
          | {if0 <CFLAE> <CFLAE> <CFLAE>}
          | {fun {<id>} <CFLAE>}
          | {<CFLAE> <CFLAE>}
```

Will add constructs for:

- changing the value of a variable
- defining a box, setting and getting the value in a box,
- sequencing two expressions

Adding State Syntactically

```
<CFLAE> ::= <num>
          | {+ <CFLAE> <CFLAE>}
          | {- <CFLAE> <CFLAE>}
          | {* <CFLAE> <CFLAE>}
          | {let {<id> <CFLAE>} <CFLAE>}
          | <id>
          | {if0 <CFLAE> <CFLAE> <CFLAE>}
          | {fun {<id>} <CFLAE>}
          | {<CFLAE> <CFLAE>}
```

```
| {newbox <SCFLAE>}
| {setbox <SCFLAE> <SCFLAE>}
| {openbox <SCFLAE>}
| {set <id> <SCFLAE>}
| {seqn <SCFLAE> <SCFLAE>}
```

Adding State Semantically

- How do we implement these constructs?

```
<SCFLAE> ::=
            ...
            | {newbox <SCFLAE>}
            | {setbox <SCFLAE> <SCFLAE>}
            | {openbox <SCFLAE>}
            | {set <id> <SCFLAE>}
            | {seqn <SCFLAE> <SCFLAE>}
```

- Two choices:
  - Meta interpreter: Using Scala boxes and variables
  - Syntactic interpreter: By implementing a mutation-free interpreter.

- Which one do you prefer?

Reflecting on Semantics of Sequences

A first try to implement **`seqn`** expressions following the pattern that we have used so far...

```
def interp(expr: SCFLAE, env: Env): Val = expr match {
  case Seqn(e1, e2) => {
    interp(e1, env)
    interp(e2, env)
  }
}
```

What do you think?
Will this work?

Reflecting on Semantics of Sequences

Quiz: What should these codes evaluate to? 0 or 1?

```
val box = Box(0)
box.value = 1 + box.value
box.value
```

```
var b = 0
b = 1 + b
b
```

```
Let("b", NewBox(0),
  Seqn(
    SetBox("b", Add(1, OpenBox("b"))),
    OpenBox("b")))
```

```
Let("b", 0,
  Seqn(
    SetId("b", Add(1, "b")),
    "b"))
```

Reflecting on Semantics of Sequences

```
val box = Box(0)
box.value = 1 + box.value
box.value
```

```
var b = 0
b = 1 + b
b
```

```
Let('b, NewBox(0),
  Seqn(
    SetBox('b, Add(1, OpenBox('b))),
    OpenBox('b)))
```

```
Let("b", 0,
  Seqn(
    SetId("b", Add(1, "b")),
    "b"))
```

- The answer is 1.
- The mutation in the first operation in the sequence has an effect on the output of the second (which would otherwise evaluate to 0).

Reflecting on Semantics of Sequences

```
val box = Box(0)
box.value = 1 + box.value
box.value
```

```
var b = 0
b = 1 + b
b
```

What would they evaluate to with the semantics implemented in our current interpretation?

```
def interp(expr: SCFLAE, env: Env): Val = expr match {
  case Seqn(e1, e2) => {
    interp(e1, env)
    interp(e2, env)
  }
}
```

Reflecting on Semantics of Sequences

```
def interp(expr: SCFLAE, env: Env): Val = expr match {
  case Seqn(e1, e2) => {
    interp(e1, env)
    interp(e2, env)
  }
}
```

- This will not work, because there is no way the interpretation of the first sequent can affect the interpretation of the second.
- Somehow we need to **"chain" interpretation effects**.

How can we do that?

Adding State Semantically

Modeling state and mutation requires a new interpretation style ...

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A First Idea for Chaining Effects

Use environments that flow through interpreter calls as the medium for chaining the mutation effects.

Using Environments for Chaining Effects

- The interpreter returns both the value of an expression and an updated environment reflecting mutation effects.

- **`interp`** returns a tuple **`Value × Environment`**

Using Environments for Chaining Effects

The interpretation of **`seqn`** uses the environment resulting from evaluating the first sequent to interpret the second.

`(interp e1 ...)` returns a modified environment

`(interp e2 ...)` uses updated environment.

```
def interp(expr: SCFLAE, env: Env): (Val, Env) = expr match {
  //...
  case Seqn(e1, e2) => {
    val (res, env1) = interp(e1, env)
    interp(e2, env1)
  }
}
```

What do you think?
Will this work?

Using Environments for Chaining Effects

- Consider the following code.

```
Seqn(
   Let("x", 10, "x"),
   "x")
```

- What would you expect it to evaluate to?
- What would it actually evaluate to?

A Evaluates to 10

B Throws error

C Undefined behavior

Using Environments for Chaining Effects

```
Seqn(
   Let("x", 10, "x"),
   "x")
```

The new “environment-passing style” of interpretation tells that **`x`** in the second expression in sequence will be evaluated in an environment that binds **`x`** to **`10`**.

This contradicts static scoping, which tells us that **`x`** in the second expression is unbound.

Using Environments for Chaining Effects

```
Seqn(
   Let("x", 10, "x"),
   "x")
```

We will have to remove bindings introduced by **`let ...`** complicates the implementation...

But, even if we are willing to work out the details, does this really solve the problem?

Using Environments for Chaining Effects

- Consider the following code.

```
val a = Box(1)
def f(x: Int) = x + a.value
a.value = 2
println(f(5))
```

```
Let("a", NewBox(1),
 Let("f", Fun("x", Add("x", OpenBox("a"))),
   Seqn(
     SetBox("a", 2),
     App("f", 5))))
```

- What would you expect it to evaluate to?
- What would it evaluate to using environment-passing style?

Using Environments for Chaining Effects

```
val a = Box(1)
def f(x: Int) = x + a.value
a.value = 2
println(f(5))
```

Would like the effect of the first sequent of the seqn block to be visible at function application.

Would our idea of chaining state changes via environments provide this expected/desired behavior? If no, why not?

Using Environments for Chaining Effects

```
val a = Box(1)
def f(x: Int) = x + a.value
a.value = 2
println(f(5))
```

- Even if we pass the **updated environment** to the interpretation of the application, it **will be ignored** since `f` **closes the body in the environment at definition time**.
- We would have to use the environment at application time, which introduces dynamic scoping!

Static scoping rules for function application will defeat environment-passing style.

Bottom Line

Using environments for chaining effects will not work!

Any other idea?

The Solution: Introducing the Store

We need **two repositories of information**:

- The **environment** for statically scoped bindings.
- The **store** for tracking dynamic changes.

- **Environment** maps **identifiers to locations** in the store.
- **Store** maps **locations to values**.

- **Identifier lookup** becomes a **two-stage process**

- A possible implementation: Use Scala maps

```
type Env = Map[Symbol, Location]
type Store = Map[Location, Val]
```

Store-Passing Interpretation Style

Will use a **modified store to chain the interpretation** of expressions. Interpreter returns a tuple **`Value × Store`**

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store)
```

This style of passing the current store in and the updated store out of every expression's evaluation is called **store-passing style**.

The Interpreter

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store)
```

Implementing Sequencing

With store-passing style, implementing **`seqn`** is straightforward

```
case Seqn(e1, e2) => {
  val (v1, store1) = interp(e1, env, store)
  interp(e2, env, store1)
}
```

Terms that are Syntactically Values

- Terms that are already syntactically values do not affect the store (since they require no further evaluation).

- They return the store unaltered.

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
  // ...
  case Num(n) => (NumV(n), store)
  case Fun(arg, body) => (Closure(arg, body, env), store)
  // ...
  case Id(name) => (store(env(name)), store)
  // ...
}
```

Note the use of functional environments…

Next: Interpreting Arithmetic Expressions and Function Application.

Anything special to consider?

Left or Right?

When it comes to arithmetic and function applications expressions, we have to decide on the order of evaluation.

```
Let("b", 4,
   Add("b",
     Seqn(
       SetId("b", 5),
     "b")))
```

With left-to-right evaluation the result is 9.

With right-to-left evaluation the result is 10.

- So far, we delegated decisions about the order of evaluating sub-expressions to the base language.
- Store-passing interpretation forces us to make this decision ourselves.

Left or Right?

This decision is a semantic one.

- Without mutations, one branch of an addition could not affect the value produced by the other branch.
- Now, each branch can have mutations that impact the value of the other.

We must choose some order so that programmers can predict what a program is going to do!

We will use a left-to-right and function-before-argument order of execution.

Arithmetic Expressions

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
  // ...
  case Add(lhs, rhs) => {



  }
}
```

Arithmetic Expressions

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
  // ...
  case Add(lhs, rhs) => {
    val (lhsv, store1) = interp(lhs, env, store)




  }
}
```

Arithmetic Expressions

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
  // ...
  case Add(lhs, rhs) => {
    val (lhsv, store1) = interp(lhs, env, store)
    lhsv match {
      case NumV(n1) => {




    }
  }
}
```

Arithmetic Expressions

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
  // ...
  case Add(lhs, rhs) => {
    val (lhsv, store1) = interp(lhs, env, store)
    lhsv match {
      case NumV(n1) => {
        val (rhsv, store2) = interp(rhs, env, store1)
        rhsv match {
          case NumV(n2) => (NumV(n1 + n2), store2)
          case _ => sys.error(
            "can only add numbers, but got right %s".format(rhsv))
        }
      }
      case _ => sys.error(
        "can only add numbers, but got left '%s'".format(lhsv))
    }
  }
}
```

Next:
Function Application

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
  ...
  case App(funExpr, argExpr) => { ... }
}
```

What are the steps?

Function Application

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
  ...
  case App(funExpr, argExpr) => { ... }
}
```

1. Interpret **`funExpr`** in **`env`** and **`store`** (function-before-arg order), yielding **`funValue`** (a closure) and **`funStore`**.

2. Interpret **`argExpr`** in **`env`** and **`funStore`**, yielding **`argValue`** and **`argStore`**.

3. Interpret **`funValue`**'s body in the following:
   1. Store: Reserve a new location in **`argStore`** (**`newLoc`**) and put **`argValue`** to it.
   2. Environment: Extend **`funValue`**'s environment with a binding for **`funValue`**'s argument to **`newLoc`**.

Function Application

```
def interp(expr: SCFLAE, env:Env, store: Store): (Val, Store) = expr match {
 ...
case App(funExpr, argExpr) => {
  val (funV, funStore) = interp(funExpr, env, store)
  val (argV, argStore) = interp(argExpr, env, funStore)

  funV match {

    case Closure(fParam, fBody, fEnv) => {
      val newLoc = nextLocation
      interp(fBody, fEnv + (fParam -> newLoc), argStore + (newLoc -> argV))
    }

    case _ => sys.error("can only apply functions, but got: " + funV)
   }
 }
}
```

Interpreting `if0` Conditionals

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
  // ...
  case If0(test, truth, falsity) => {
    val (res, _) = interp(test, env, store)
    res match {
      case NumV(n) => {
        if (n == 0) interp(truth, env, store)
        else interp(falsity, env, store)
      }
      case _ => sys.error("can only test numbers, but got: " + res)
  }
}
```

What about this interpretation of `if0` expressions?

```
Let("b", 0,
   If0(Seqn(SetId("b", 5), 5),
      1, "b"))
```

???

Interpreting `if0` Conditionals

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
 // ...
 case If0(test, truth, falsity) => {
  val (res, store1) = interp(test, env, store)
  res match {
   case NumV(n) => {
    if (n == 0) interp(truth, env, store1)
    else interp(falsity, env, store1)
   }
   case _ => sys.error("can only test numbers, but got: " + res)
  }
}
```

- Need to use the modified store returned by the interpretation of `test` in the interpretation of the conditional branches.
- This allows to propagate possible mutations in `test`.

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Adding Variables

```
<SCFAE> ::= ...
            | {set <id> <SCFAE>}
```

Implementing Variables

- For evaluating **`set`**, the identifier is not fully evaluated.
- We just retrieve the identifier's location from the environment.

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) =
expr match {
 …
 case SetId(id, valExpr) => {
   val (value, s1) = interp(valExpr, env, store)
   val loc = env(id)
   (value, s1 + (loc -> value))
 }
}
```

L-Values and R-Values

- Variables in an assignment position (e.g., left of := or =), are only **partially resolved**.
  - These special values (the location of an identifier) are called **`l-values`**
  - Only appear on the left-hand side of assignments.

- Variables in expressions are **fully resolved** to a value (by store-lookup and env-lookup). They are called **`r-values`**.

Quiz: Which variables are l-values, which are r-values?

```
x = 2;
y = x;
```

Variables and Function Application

Are function application and variables independent language features?

Variables and Function Application

```
Let("v", 0,
   Let("f", Fun("x", SetId("x", 5)),
     Seqn(
       App("f", "v"),
       "v")))
```

What does the program evaluate to?

It could be 0 or 5.
Both are reasonable depending on the parameter passing semantics, i.e., whether functions are passed r-values or l-values (locations).

Variables and Function Application

```
Let("v", 0,
  Let("f", Fun("x", SetId("x", 5)),
    Seqn(
      App("f", "v"),
      "v")))
```

What does the program evaluate to with our interpreter?

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
 ...
 case App(funExpr, argExpr) => {
  val (funV, funStore) = interp(funExpr, env, store)
  val (argV, argStore) = interp(argExpr, env, funStore)
  funV match {
   case Closure(fParam, fBody, fEnv) => {
    val newLoc = nextLocation
    interp(fBody, fEnv + (fParam -> newLoc), argStore + (newLoc -> argV))
   }
   case _ => sys.error("can only apply functions, but got: " + funV)
  }
 }
}
```

Variables and Function Application

```
Let("v", 0,
   Let("f", Fun("x", SetId("x", 5)),
     Seqn(
       App("f", "v"),
       "v")))
```

Here the answer is 0

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
 ...
 case App(funExpr, argExpr) => {
  val (funV, funStore) = interp(funExpr, env, store)
  val (argV, argStore) = interp(argExpr, env, funStore)
  funV match {
    case Closure(fParam, fBody, fEnv) => {
      val newLoc = nextLocation
      interp(fBody, fEnv + (fParam -> newLoc), argStore + (newLoc -> argV))
    }
    case _ => sys.error("can only apply functions, but got: " + funV)
  }
 }
}
```

Call-by-Value Eager Evaluation

- **`argValue`** is put into a **new store location** and the formal parameter is bound to that new location in the environment.
- **Changes to the formal parameter** in the body of the function **are not visible outside the function!**

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
 ...
 case App(funExpr, argExpr) => {
  val (funV, funStore) = interp(funExpr, env, store)
  val (argV, argStore) = interp(argExpr, env, funStore)
  funV match {
   case Closure(fParam, fBody, fEnv) => {
    val newLoc = nextLocation
    interp(fBody, fEnv + (fParam -> newLoc), argStore + (newLoc -> argV))
   }
   case _ => sys.error("can only apply functions, but got: " + funV)
  }
 }
}
```

Standard form of eager evaluation called "call-by-value"

Call-by-Reference Eager Evaluation

- A reference to (the location of) the actual argument (not its value) is passed when a function is called.

- Updates to the reference (location) within the called function become visible to the calling context.

Quiz: What does the following program evaluates to with call-by-reference semantics?

```
Let("v", 0,
   Let("f", Fun("x", SetId("x", 5)),
     Seqn(
       App("f", "v"),
       "v")))
```

Call-by-Reference Functions

- We introduce call-by-reference functions to our language

- Such functions are syntactically the same as call-by-value functions except for the different introducing keyword (`refun` instead of `fun`).

```
<SCFLAE> ::= ...
           | {refun {<id>} <SCFLAE>}
```

Implementing Call-by-Reference

- Like a **`fun`**, a **`refun`** also evaluates to a closure.

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
  // ...
  case ReFun(arg, body) => (RefClosure(arg, body, env), store)
}
```

- **`RefClosure`** values have the same fields as **`Closure`** values.

- Different tags are used to distinguish between the two kind of functions at application time.

Application with Call-by-Reference

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
 // ...
 case App(funExpr, argExpr) => {
  val (funV, funStore) = interp(funExpr, env, store)
  funV match {

  // case Closure(fParam, fBody, fEnv) => ...

  case RefClosure(fParam, fBody, fEnv) => argExpr match {
   case Id(argName) => {
    interp(fBody, fEnv + (fParam -> env(argName)), funStore)
   }
   case _ => sys.error("can only call identifiers by reference")
  }
  case _ => sys.error("can only apply functions, got: " + funV)
 }
}
```

store is not updated

assumes **`argExpr`** to be a syntactic identifier.

location of actual argument bound to formal parameter.

Reflecting on Call-by-Reference

Advantage:
Less allocation to the store.

Disadvantage:
Creates aliases, making reasoning about programs harder.

C allows to mark certain parameters of a multi-parameter procedure as **&** (other languages as **ref**).

Call-by-Value or Call-by-Reference

To master complexity, we need **abstraction boundaries** between units of modularity (e.g., procedures), that **limit hidden interactions** and **restrict interferences** to facilitate reasoning about their behavior in isolation.

There is agreement in modern languages that parameters should be passed by values!

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What are Boxes?

- Do we understand the semantics of boxes to start with?
- Quiz: What does the following evaluate to?

```
class Box[A](var value: A)
val b1 = new Box(5)
val b2 = new Box(5)
val b3 = b1

b1 == b2

b1.value = 6

b1.value
b2.value
b3.value
```

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```

```
> false
  6
  5
  6
```

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b1.value
b2.value
b3.value
```

```
> false
  6
  5
  6
```

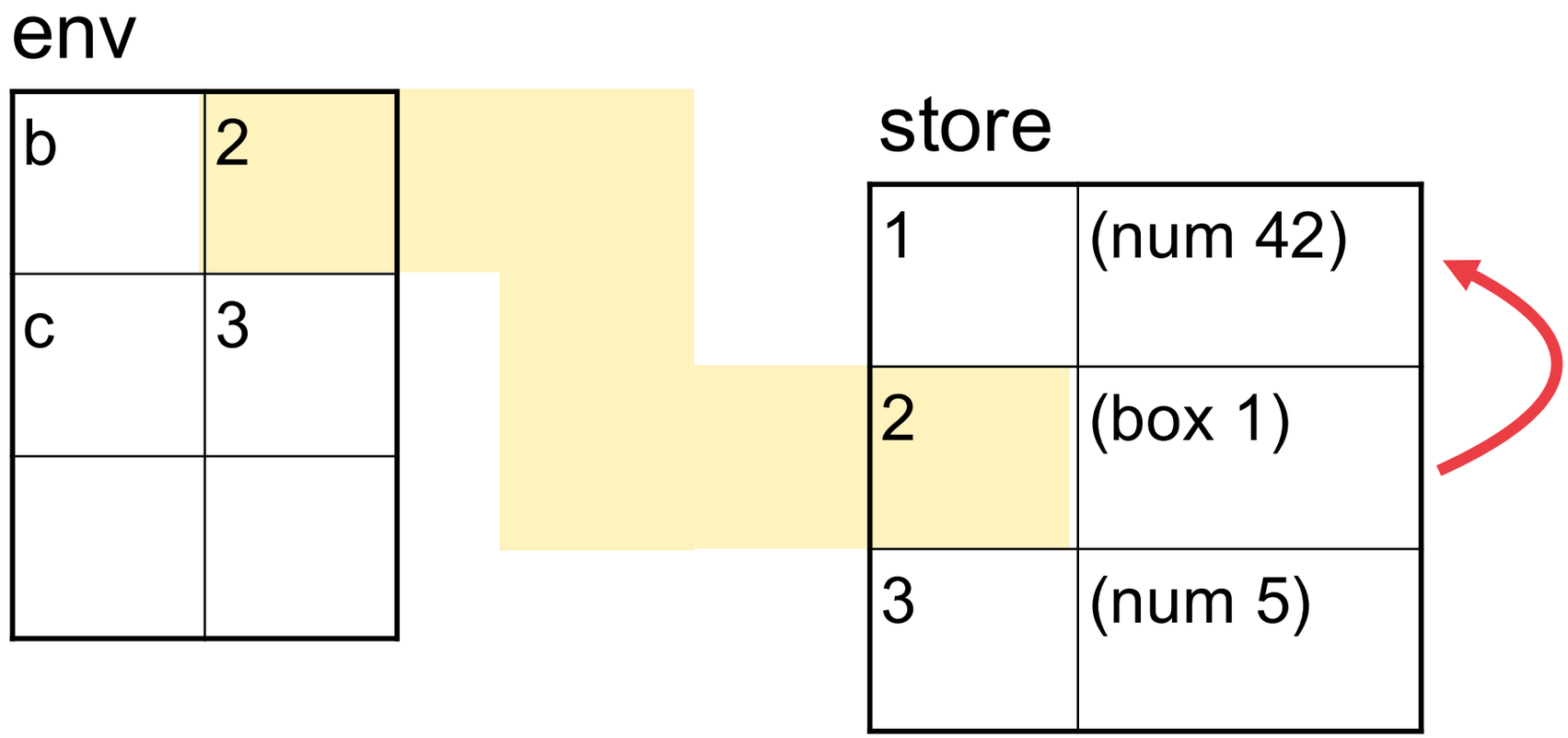
- b1 and b2 contain the same values but are not the same. Equality versus identity.
- One can think of b1 and b2 as two different places that contain the same values.
- b3 is defined to be the same place as b1 and hence is identical to b1.

Boxed Values

- Boxes are a third kind of values.

```
sealed abstract class Val
case class NumV(n: Int) extends Val
case class Closure(param: Symbol, body: SCFLAE, env: Env) extends Val
case class Box(location: Location) extends Val
```

- **A box value is a location in the store.**
- At this location, the content of the box is stored.



Creating Mutable Boxes

1. Interpret **`valueExpr`** resulting in a **`value`** and potentially updated store, **`boxStore`**.
2. Get new location in the most recent store.
3. Put value into new location and store this location in a box value.
4. Return the box value and the updated store.

```
def interp(expr: SCFLAE, env:Env, store:Store): (Val, Store) = expr match {
  //...
  case NewBox(valueExpr) => {
    val (value, boxStore) = interp(valueExpr, env, store)
    l newLoc = nextLocation
    ox(newLoc), boxStore + (newLoc -> value))
  }
}
```

1 2 3 3 4

Creating Mutable Boxes by Example

```
(let (b1 (newbox 5))
  (let (b2 (newbox 5))
       (... some-expr ...)
```

```
interp (newbox 5) [] [] ~> ((box 1),
```

1	(num 5)

```
)
```

```
interp (let b1 (newbox 5)) []
```

1	(num 5)

```
)
```

```
~>
```

env

b1	2

store

2	(box 1)
1	(num 5)

Creating Mutable Boxes by Example

```
(let (b1 (newbox 5))
  (let (b2 (newbox 5))
       (... some-expr ...)
```

`interp (newbox 5)`

env	
b1	2

store	
2	(box 1)
1	(num 5)

`~> ((box 3),`

store	
3	(num 5)
2	(box 1)
1	(num 5)

`)`

`interp (let b2 (newbox 5))`

`~>`

env	
b2	4
b1	2

store	
4	(box 3)
3	(num 5)
2	(box 1)
1	(num 5)

env	
b1	2

store	
3	(num 5)
2	(box 1)
1	(num 5)

Creating Mutable Boxes by Example

```
(let (b1 (newbox 5))
  (let (b2 (newbox 5))
       (... some-expr ...))
```

interp some-exp

env	
b2	4
b1	2

store	
4	(box 3)
3	(num 5)
2	(box 1)
1	(num 5)

Opening a Box

- Evaluate the box expression to a box value, **`boxV`**, and a potentially updated store, **`s1`**.
- Return the result of looking up **`s1`** at the location in the box value, **`s1(loc)`**, and **`s1`**.

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {
 // ...
 case OpenBox(boxExpr) => {
  val (boxV, s1) = interp(boxExpr, env, store)
  boxV match {
   case Box(loc) => (s1(loc), s1)
   case _ => sys.error("can only open boxes, but got: " + boxV)
  }
 }
}
```

Opening a Box by Example

```
interp (openbox b1)
```

env	
b2	4
b1	2

store	
4	(box 3)
3	(num 5)
2	(box 1)
1	(num 5)

- The composed env and store lookup triggered by the interpretation of `b1` returns `(box 1)`.
- Return the value in the location enclosed in `(box 1)` - `(num 5)` - and whatever the most recent store is.

Setting a Box

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {

  //...
  case SBox(boxExpr, valueExpr) => {





  }
}
```

Setting a Box

- Evaluate the box expression to a box value, **`boxV`**, and a potentially updated store, **`s1`**.

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {

  //...
  case SBox(boxExpr, valueExpr) => {
    val (boxV, s1) = interp(boxExpr, env, store)





  }
}
```

Setting a Box

- Evaluate the box expression to a box value, **`boxV`**, and a potentially updated store, **`s1`**.
- Evaluate **`valueExpr`** to a **`value`** and a store **`s2`**

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {

  //...
  case SetBox(boxExpr, valueExpr) => {
    val (boxV, s1) = interp(boxExpr, env, store)
    val (value, s2) = interp(valueExpr, env, s1)




  }
}
```

Setting a Box

- Evaluate the box expression to a box value, **`boxV`**, and a potentially updated store, **`s1`**.
- Evaluate **`valueExpr`** to a **`value`** and a store **`s2`**
- Update **`s2`** with a binding of the location in **`boxV`** to **`value`**
- Return the new **`value`** and the updated **`s2`** store.

```
def interp(expr: SCFLAE, env: Env, store: Store): (Val, Store) = expr match {

  //...
  case SetBox(boxExpr, valueExpr) => {
    val (boxV, s1) = interp(boxExpr, env, store)
    val (value, s2) = interp(valueExpr, env, s1)
    boxV match {
      case Box(loc) => (value, s2 + (loc -> value))
      case _ => sys.error("can only set to boxes, but got:" + boxV)
    }
  }
}
```

Setting a Box by Example

`interp (setbox b1 6)`

env	
b2	4
b1	2

store	
4	(box 3)
3	(num 5)
2	(box 1)
1	(num 5)

`~> (num 6)`

store	
1	(num 6)
4	(box 3)
3	(num 5)
2	(box 1)
1	(num 5)

- The composed env and store lookup triggered by interpretation of `b1` returns `(box 1)` and leaves store unchanged.
- Interpretation of `6` returns `(num 6)` and leaves store unchanged.
- The result is `(num 6)` and the store extended with a binding of the location enclosed in `(box 1)` to `(num 6)`

Getting a New Location in a Store?

- How can we implement next-location?

```
def nextLocation: Location
```

- What about this implementation?

```
type Location = Int
var _currentLocation = 0
def nextLocation: Location = {
  _currentLocation += 1
  _currentLocation
}
```

Unsatisfying, because it relies on a variable - wanted a syntactic mutation-free interpreter. How can we get rid of the box?

Outline

- State - in Scala and in our languages
- An interpretation style for modeling the effect of state
- Variables
- Mutable data structures (Boxes)
- Summary

Store-passing Interpretation Style

- Supporting state requires a new interpretation style to enable propagation of effects: **Store-passing interpretation**

- Store as a global medium for keeping track of dynamic value updates.

- **Static scope of variables** versus **dynamic extent of values**.
  - Environments as repositories for static scope.
  - Stores as repositories for dynamic extent.

Variables versus Mutable Data Structures

- **Variable**: **Location** corresponding to a **variable** is associated with a **"real" value**.

- **Mutable Data Structure**: **Location** corresponding to an identifier is associated with **a special "box value"** - denoting the **location where the mutable data is stored**.

Variables versus Mutable Data Structures

- `set`
  - Requires the first sub-expression to be an identifier.
  - Changes the binding of the identifier in the store.

- `set-box`
  - Allows any expression in the first sub-expression.
  - Must reduce to a “box value”.
  - The location referred to by the box value gets associated with a new “real” value.

Call-by-Value versus Call-by-Reference

- **Variables and function application are not independent language features.**

- When variables are supported, two (eager) evaluation semantics are possible: **call-by-value & call-by-reference**.

- Call-by-reference is problematic and is not supported in many languages.

Scope versus Extent

- Distinction between environment and store reveals different patterns of flow.
- Closures capture the environment at function definition, but not the store.
- The store is a global record of changes made during execution.

Scope versus Extent

- The two flows of values through the interpreter reflect the difference between names (identifiers) and values.

- Values remain in the store, even if the name that introduced them disappears from the environment.

- Not inherently a problem, since another name may have been associated with the value in the flow of computation.

- **Identifiers have lexical scope**.
- **Values** have (potentially indefinite) **dynamic extent**.

Issues With Confusing Scope and Extent

- Some languages confuse lexical scope and dynamic extent.
- This is a common problem in languages like C and C++.
- When an identifier ceases to be in scope, they remove the value corresponding to the identifier.
  - If some other identifier may still have a reference to it, this premature removal of the value will inevitably lead to a system crash.
  - Depending on how the implementation “removes” the value, this may lead to extremely vexing bugs.

Issues With Confusing Scope and Extent

```
{
    char *dp = NULL;
    /* ... */
    {
        char c;
        dp = &c;
    }
      /* c falls out of scope */
        /* dp is now a dangling pointer */
}
```

The stack is destroyed after a frame returns. But the pointer is not set to NULL again.

Severe security implications: The Heartbleed bug (2014) in OpenSSL was partly caused by dangling pointers

When are Values Removed from Store?

- Right now, never.

- **Garbage collection** lets languages **dissociate scope from the reclamation of space** consumed by values.
  - Can remove unreachable values.
  - Can be quite performant